

BigchainDB: A Popular and Powerful Database *for* Blockchain

Blockchain technology offers decentralised control, where no one has any ownership or control over the network; and a tamper-proof environment, where nothing gets deleted from network storage. It also has the ability to create and transfer assets on the network without reliance on a central entity. Blockchain storage, however, requires a powerful database, and this is where BigchainDB comes in.



As single monolithic blockchain technologies now have a wide range of applications, they have been re-framed and refactored into building blocks at four levels of the stack:

- Applications
- Decentralised computing platforms
- Decentralised storage (ledgers, file systems, databases), decentralised processing (smart contracts) and decentralised communication

- Cryptographic primitives, consensus protocols and other algorithms
- Blockchain storage requires a powerful database, and the traditional blockchain as a database (DB) is pathetic, as the throughput is just a few transactions per second (tps), latency before a single confirmed write is 10 minutes, and the capacity is a few dozen gigabytes. Also, adding new nodes causes problems in terms of network

traffic, congestion, latency, throughput and capacity. As compared to traditional databases, a modern distributed database provides throughput exceeding 1 million tps, 1000 TB storage, latency that is a fraction of a second, with no effect on throughput and latency when increasing the number of nodes. In addition, modern databases have good capabilities for insertion, querying, access control and security.